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LCA FOR ENERGY SYSTEMS AND FOOD PRODUCTS

Comparative life cycle assessment: environmental impact of beef jerky and its plant-based analogue

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Abstract

Purpose Rising population, income, and urbanisation are increasing demand for animal protein, intensifying environmental pressures associated with meat production. This study evaluates the environmental performance of a plant-based jerky analogue compared with conventional beef jerky, assessing whether a product derived from plant protein blends via high-moisture extrusion can reduce impacts across multiple environmental impact categories.

Methods A cradle-to-gate life cycle assessment (LCA) was performed using the Environmental Footprint 3.1 method to compare beef jerky with a plant-based jerky analogue derived from a plant protein blend. System boundaries covered raw material production, processing, and packaging. The beef jerky inventory was compiled from scientific literature, while primary data from the manufacturer were used for the plant-based product. Background data were sourced from ecoinvent v3.10, Agri-Footprint v6.3, and AGRIBALYSE v3.1 databases. Environmental impacts were assessed across 16 impact categories for two functional units: 1 kg of product and 1 kg of protein. Contribution analysis identified key drivers, and sensitivity and uncertainty analysis evaluated the robustness of the results.

Results and discussion Plant-based jerky showed lower environmental impacts in 15 out of 16 categories for both functional units, with ionising radiation being the only exception. On a product basis, reductions reached 95.0% for climate change and 49.0% for water use. The land-use impact for the plant-based product was negative, which was attributed to the selected characterization method rather than treating it as an absolute environmental benefit. Raw material production was the dominant contributor in both systems. For beef jerky, livestock production drove most impacts, while crop cultivation and protein extraction were the main contributors for the plant-based alternative. Results remained consistent under different sensitivity assessments, confirming robustness. Additional processing stages did not offset the overall environmental advantages of the plant-based product.

Conclusions The plant-based jerky demonstrates substantially lower environmental impacts than beef jerky across most impact categories, indicating strong potential as a more sustainable alternative. In both systems, raw material production dominates environmental burdens, highlighting the need for targeted upstream improvements to achieve further reductions.

Recommendations Future research should evaluate the combined usage of multiple plant protein sources and account for the cumulative impacts of additional processing stages. Comparative assessments should also incorporate nutritional factors, including protein digestibility, bioavailability, and conversion efficiency, to enable a more comprehensive evaluation of plant- and animal-based products.

Keywords Life cycle assessment (LCA) · Plant-based analogue · Beef jerky · Plant proteins · Environmental footprint · Sustainability · Environmental impacts

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1 Introduction

In February 2026, Fortune magazine reported on raising popularity of meat snack products among US consumers. Within the savoury snacks category, meat snacks have shown strong sales performance, with a compound annual growth rate of 6.6% from 2020 to 2025. This market trend, often referred to as the "Ozempic effect", is likely linked to the widespread use of glucagon-like peptide-1 (GLP-1) medications, which has increased consumer interest in balanced, protein-rich diets (Lichtenberg 2026). Meat snack products, such as biltong and jerky, are commonly consumed between meals to satisfy short-term hunger and provide energy and nutrients, making them a convenient option for preserving muscles during a low-calorie diet (Lichtenberg 2026). However, many commercially available products require more than 200 g of raw beef to produce 100 g of dried beef, making it a significant contributor to food-related environmental impacts (Clark et al. 2022).

In general, livestock farming plays a fundamental role in global food security, supplying essential nutrition to countless families and societies (Food and Agriculture Organization of the United Nations (FAO) 2023). However, this sector contributes substantially to greenhouse gas (GHG) emissions, with cattle production accounting for the largest share: 3.8 Gt CO_2eq, or 62% of total livestock-related emissions (FAO 2023). With the continuously growing global demand for animal protein, GHG emissions could reach 9.1 Gt CO_2eq by 2050 (FAO 2023), further intensifying extreme climate events such as heatwaves, droughts, tropical cyclones, and storms (Intergovernmental Panel on Climate Change (IPCC) 2023). Beyond the environmental concerns, high consumption of red meat and processed meat has been linked to an increased prevalence of non-communicable diseases such as heart diseases, cancer, and diabetes (Wepner et al. 2023). In this context, primary production and food processing are considered key sectors with significant responsibility for addressing those environmental and public health challenges (Wepner et al. 2023).

To meet the evolving demand for nutrient-dense and sustainable diets, the food industry is increasingly shifting its focus toward alternative protein sources, such as plants, algae, and fungi-based alternatives for beverages, meat products, and snacks (Sharif et al. 2022). In recent years, the plant-based meat market has expanded substantially, offering a growing number of innovative products, with plant-based jerky showing strong growth potential within this sector (Ignaszewski 2023). To mimic the appearance and texture of conventional meat, vegan meat typically achieves a fibrous structure by texturizing globular plant proteins, most commonly using extrusion technologies. (Li et al. 2022). Extrusion refers to a process in which a liquid or semi-liquid mass is forced through a die with a defined cross-section to form structured products (Kazemzadeh 2016).

Many studies suggest that plant-based food products may be promising, more sustainable alternatives to their animal-based counterparts (Detzel et al. 2021; Heller et al. 2025; Saerens et al. 2021). In general, beef is consistently associated with substantially higher environmental burdens than most ingredients used in meat analogues (Smetana et al. 2023). Nonetheless, the environmental performance of meat alternatives varies and largely depends on the choice of raw materials. Among current options, soy-based analogues show some of the lowest environmental impacts, mainly driven by energy use during frying (58%) and soybean processing (18%) (Smetana et al. 2015). In general, crop processing contributes significantly to the overall environmental performance of plant-based alternatives. Specifically, processing crops into protein isolates is closely linked to high energy requirements and process water consumption (Detzel et al. 2021). Milder fractionation techniques, which require less energy and fewer auxiliary chemicals for protein extraction, can substantially reduce the environmental impact of plant proteins (Lie-Piang et al. 2021). However, less refinement may reduce protein purity and yield, which must be considered when selecting ingredients for food product formulations (Lie-Piang et al. 2021).

Although Clark et al. (2022) have identified dried beef products as the highest-impact food product, research on the environmental performance of plant-based counterparts remains scarce. Existing studies on plant-based jerky focus more on market research, including consumer perceptions of organoleptic properties, packaging, and price sensitivity (Chaffee and Ardoin 2025; Veniranda and Surya 2022). Because of its rising growth potential in the plant-based market, this life cycle assessment compares the environmental performance of beef jerky with a plant-based analogue produced by high-moisture extrusion. The plant-based jerky formulation consisted of multiple plant proteins, derived from pea, faba bean, pumpkin seed, and sunflower seed.

2 Materials and methods

2.1 Goal of the life cycle assessment

This life cycle assessment compared the environmental performance of beef jerky and plant-based jerky to help consumers make more sustainable choices among high-protein snack products. Additionally, a hot-spot analysis identified opportunities for product developers to reduce the ecological footprint of their processes.

This study followed international standards and used an attributional approach. The input and output flows of the product systems were related to a defined functional unit to compare the environmental performance of beef jerky and plant-based jerky (Lopes Silva 2021).

2.2 Functional unit

Both jerky products considered in this study have a similar energy content (297 kcal per 100 g of beef jerky and 307 kcal per 100 g of plant-based jerky) and are marketed as "high-protein" products. Beef jerky contains 56 g of protein per 100 g of product, while the plant-based analogue provides 39 g of protein per 100 g of product. Based on these nutritional differences, two functional units were defined to reflect the environmental impacts clearly:

Product mass-based functional unit: 1 kg of finished (packed) product

Protein- based functional unit: 1 kg of protein in the product

2.3 System boundaries

The system boundary for this study includes the unit processes of the products and their corresponding inputs and outputs that determine environmental performance. A cradle-to-gate assessment was performed, including raw material acquisition, packaging recycling, machinery requirements, raw material processing, and final product packaging. Both production systems were considered as continuous processes. Furthermore, the study excluded food waste disposal, production rejects, subsequent distribution channels, and consumer consumption because their relative contribution to overall environmental impact was assumed to be similar for both products.

The beef jerky production system (Fig. 1) begins with raw material acquisition. Generally, thin slices of whole beef muscles, typically from the top and bottom rounds of hindquarters, are used as the main ingredient for jerky production (Hasheider 2015). In recent approaches, slaughterhouse by-products or restructured meat trimmings have been used (Kumar et al. 2019). However, this study referred to the traditional production method. Therefore, fresh beef meat was considered as primary material (Tab. S 1, SM1). Its supply involved farm-level activities, slaughtering, deboning, trimming, packaging of the final product, and refrigerated transport. Moreover, the cultivation and manufacturing of additional ingredients such as vinegar, spices, and additives were included. As powdered ingredients are commonly used in industrial food formulations for convenient storage, transport, and longer shelf life (Cuq et al. 2011), soy protein hydrolysate and yeast extract were considered in powdered form. After acquiring raw materials, traditional beef jerky production continues with the following steps: slicing the meat, marinating, and drying (Kumar et al. 2019). Finally, the dried slices are cut into bite-sized pieces and packed in a polyethene multilayer bag to protect them from light and oxygen.

The production of plant-based jerky (Fig. 2) differs from its meat-based counterpart in initial production stages. Instead of beef, a blend of four protein powders (Tab. S 1, SM1), derived from peas, faba beans, pumpkin seeds, and sunflower seeds, serves as the protein source (Lammers and Komainda 2022). Plant protein production comprises two main stages: crop cultivation (sowing, growing, and harvesting) and downstream processing, including dehulling, milling, and flour fractionation for protein extraction (Sharif et al. 2022). To achieve the characteristic jerky flavour, producers add different flavours and spices to the protein blend. Moreover, liquid ingredients such as water, glycerol, tomato

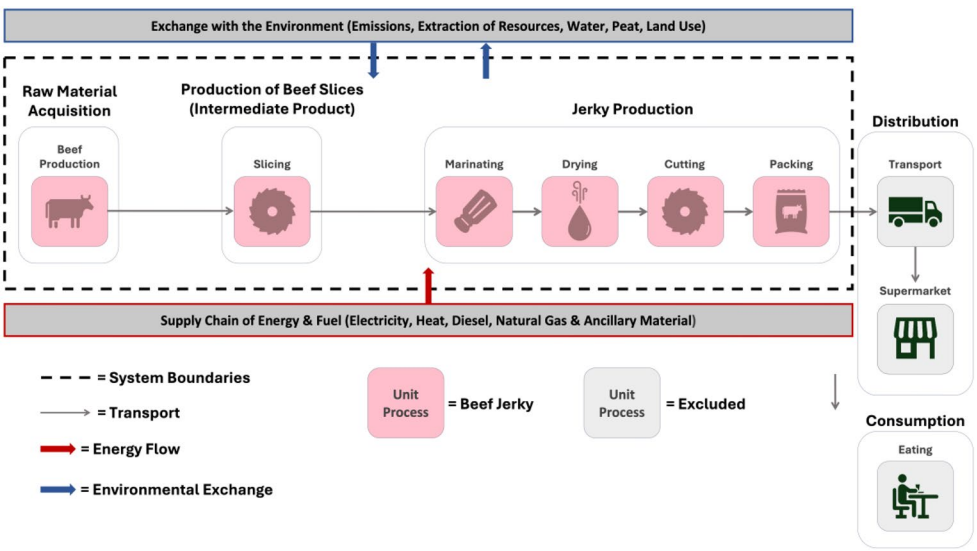


Fig. 1 System boundary of beef jerky

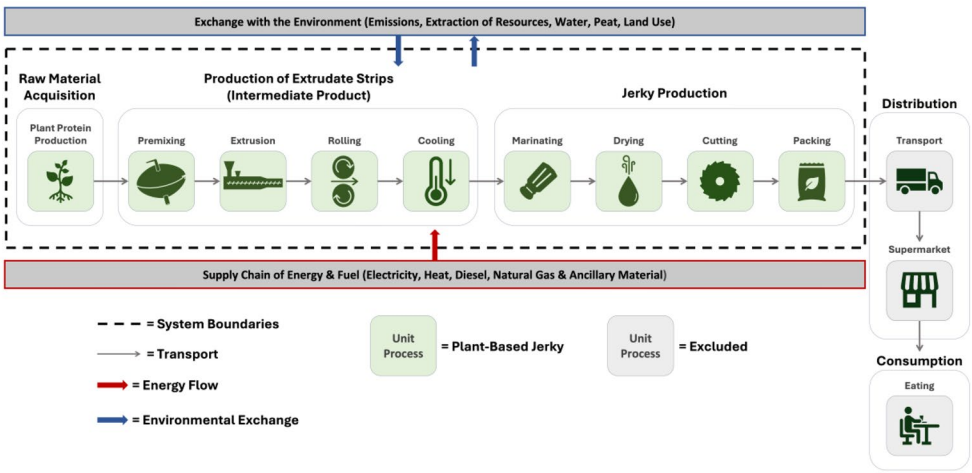


Fig. 2 System boundary of plant-based jerky

paste, and sunflower oil are part of the formulation (Lammers and Komainda 2022). The system boundary included the cultivation and manufacturing of all these ingredients and substances. After the acquisition of raw materials, plant proteins are texturized and processed to extruded strips serving as a basis for the plant-based analogue. This process is detailed in the patent DE 10 2020 210 726 A1 by Lammers and Komainda (2022). First, two mixtures are prepared. One contains all solid ingredients, including the protein powder blend, flavours, and spices. The second mixture contains all hydrophilic substances (water, glycerol, and tomato paste). Both mixtures are fed into the extruder via a solid and liquid feeder, respectively. Sunflower oil is added separately by using an oil pump. Inside the extruder barrel, the ingredients are mixed under high temperature and pressure to form a texturized product. The hot extrudate exiting the die is manually cut into long strips, which are then rolled to achieve the desired structure and thickness, and cooled on a cooling belt (Lammers and Komainda 2022). From the marinating stage onwards, the production system of plant-based jerky is identical to its meat-based counterpart. The long shelf life is mainly achieved by reducing the moisture content and water activity during drying (Kumar et al. 2019; Lammers and Komainda 2022). For both products, the same drying temperature (80 °C) was assumed. Finally, the dried extrudate strips are cut into bite-sized pieces and packed in polyethene multilayer bags to protect them from light and oxygen. A complete overview of ingredient composition (% (wt) of beef jerky and plant-based jerky is mentioned in Tab. S 1, SM1.

2.4 Life cycle inventory (LCI)

Life cycle inventory (LCI) data for the production processes of beef jerky and plant-based jerky were compiled from multiple sources, including industry data, peer-reviewed scientific literature, and three established LCI databases: ecoinvent v3.10 (Wernet et al. 2016), AGRIBALYSE v3.1 (Koch et al. 2024), and Agri-Footprint v6.3 (Durlinger et al. 2014). The complete list of inventory datasets and corresponding process assumptions is provided in the Supplementary Materials (SM) 2.

Both products contained numerous ingredients and processing steps that were not fully covered by a single LCI database. Therefore, datasets from ecoinvent v3.10 (Wernet et al. 2016), AGRIBALYSE v3.1 (Koch et al. 2024), and Agri-footprint v6.3 (Durlinger et al. 2014) were combined to develop the final inventory. Dataset selection was based on process specificity, technological relevance, geographical representativeness, and data availability. Agri-footprint v6.3 (Durlinger et al. 2014) and AGRIBALYSE v3.1 (Koch et al. 2024) were preferentially used for plant-protein ingredients and other food ingredients, while ecoinvent v3.10 (Wernet et al. 2016) was primarily used for energy, transport, general processing inputs, and as a background model for agricultural production. Methodological consistency was improved by applying common system boundaries and allocation principles across the foreground system and by selecting the cut-off system model of ecoinvent v3.10 (Wernet et al. 2016) to better align with the background modelling used in AGRIBALYSE v3.1 (Koch et al. 2024) and Agri-footprint v6.3 (Durlinger et al. 2014). Country- or region-specific datasets were selected according to the origin of each plant-protein ingredient and its processing location. When such datasets were unavailable, the most representative regional or global dataset was used. For multifunctional processes, environmental burdens were allocated among co-products according to their relative market values.

Nevertheless, combining LCI databases may introduce methodological variability because the databases can differ in activity data, agricultural emission modelling, allocation procedures, land-use modelling, and treatment of infrastructure and capital goods (Corrado et al. 2018). The infrastructure-related limitations reported for earlier versions of

Agri-footprint have been partly addressed in Agri-footprint v6.3 (Durlinger et al. 2014), which includes selected agricultural infrastructure and uses ecoinvent data for several background processes. However, capital goods are not consistently represented across all processes. These remaining differences were therefore considered as a source of uncertainty when interpreting the results.

2.4.1 Beef jerky production

Production of traditional beef jerky was modelled based on existing literature (Hasheider 2015; Kumar et al. 2019), video material (Factory Window 2024; History of Food 2024) and scientific studies by Lim et al. (2012) and Lim et al. (2014). Ingredient mass percentages were estimated from ingredient lists and nutrition information on the packaging of commercially available beef jerky products, as well as from a Korean beef jerky formula reported by Lim et al. (2012). Based on the packaging details, it was assumed that 225 g of beef is needed to produce 100 g of the finished product. Beef data was obtained from existing datasets available in Agri-footprint v6.3 (Durlinger et al. 2014) and AGRIBALYSE v3.1 (Koch et al. 2024), considering beef production in Ireland and its subsequent distribution to the manufacturing site.

2.4.2 Plant-based jerky production

Product composition of plant-based jerky was obtained directly from the manufacturer. The plant protein models were obtained either directly from existing databases or developed using comparable process models from LCI databases and the scientific literature.

2.4.2.1 Plant protein ingredients Pea protein isolate was one of the main plant protein ingredients used in the recipe. In this study, the pea protein isolate model available in Agri-footprint v6.3 (Durlinger et al. 2014) was used without further modifications. According to the soy protein concentrate production model in AGRIBALYSE v3.1 (Koch et al. 2024), soy protein concentrate is produced from soybean oil cake using wet extraction. Due to the similarity between soybeans and sunflower seeds as oil-bearing crops, a comparable processing pathway was assumed. Therefore, the soy protein concentrate model from AGRIBALYSE v3.1 (Koch et al. 2024) was adapted in combination with the sunflower seed meal production dataset from Agri-footprint v6.3 (Durlinger et al. 2014) to model the sunflower seed protein concentrate production. Data on pumpkin seed meal production were obtained by Saerens et al. (2021), who adapted the pumpkin cultivation model from the zucchini cultivation model, as both crops belong to the same family (Saerens et al. 2021). The faba bean cultivation model in ecoinvent v3.10 (Wernet et al. 2016) was selected as the background model for faba bean protein concentrate production. As faba beans and peas are both classified as legumes (Sharif et al. 2022), subsequent protein extraction stages were modelled based on an existing pea protein concentrate production model in Agri-footprint v6.3 (Durlinger et al. 2014). The mass yields of the dehulling and fractionation stages were adapted to information provided by Krenz et al. (2024) and Dumoulin et al. (2021), respectively.

2.4.2.2 Other ingredients Liquid ingredients, spices, and additives production data were obtained from the aforementioned LCI databases and modified where necessary.

2.4.3 Foreground processes

Process parameters and technical data for extrusion were provided by the manufacturer. However, primary data on energy consumption based on direct measurements were unavailable. Therefore, energy requirements were estimated using data from the technical data sheets of machines. Machine weights reported in the technical data sheets were used to estimate material requirements (stainless steel) for machine production. The same approach was applied to estimate the energy and material inputs for all foreground processes in both production lines.

R-407 C was used as the refrigerant in the extruder cooling section. As equipment-specific leakage measurements were unavailable, refrigerant emissions were estimated using an annual operational leakage rate of 8% of the installed refrigerant charge. This assumption is consistent with the operational loss rate reported for industrial refrigeration systems by ICF International (2011) and lies within the IPCC default range of 7–25% yr^{-1} for industrial refrigeration systems, including food-processing applications (Ashford et al. 2006). Liquid nitrogen was used as the cryogenic fluid in the cooling-belt system following extrusion and rolling. The quantity of liquid nitrogen required to produce 1 kg of plant-based jerky was included as an input to the cooling process together with the associated energy and material inputs.

According to packaging information, smoke flavour was added to both products. In principle, liquid smoke is a common ingredient in many jerky formulations, giving it its characteristic flavour. Additionally, the usage of smoke flavours is considered to be safer than using real smoke, as carcinogenic compounds are filtered out (Hasheider 2015). Data for liquid smoke production were obtained from Smetana and Toepfl (2015).

Based on manufacturer information, drying was performed in a smoking chamber. For beef jerky, the packaging indicated that approximately 225 g of beef were required to produce 100 g of dried product. After accounting for the marination and smoke flavour, the model assumed the evaporation of 1.389 kg of water per kg of final beef jerky. For the plant-based jerky, manufacturer data indicated a mass reduction of approximately 33% during drying, corresponding to the evaporation of 0.495 kg of water per kg of final product. Drying-energy requirements were estimated from the sensible heat required to increase the product temperature from 25 °C to 80 °C and the latent heat required for water evaporation. After accounting for an assumed dryer efficiency of 80% and a 10% heat-loss correction, the final drying-energy requirements were calculated as 1.3802 kWh kg^{-1} of beef jerky and 0.5223 kWh kg^{-1} of plant-based jerky. Detailed equations, thermophysical properties, and intermediate calculations are provided in Supplementary Materials (SM) 1; Calculation of heating energy requirement.

2.5 Life cycle impact assessment

The life cycle assessment was performed using SimaPro 9.6.0.1 software (PRé Sustainability B.V., Amersfoort, The Netherlands). The Environmental Footprint (EF) 3.1 method, developed by the European Commission, includes the Product Environmental Footprint (PEF) and the Organization Environmental Footprint (OEF), and provides reliable information on the environmental performance of products and organizations (European Commission 2021). Therefore, it was selected to assess the life cycle environmental impacts. All 16 midpoint impact assessment categories in EF 3.1 were chosen to compare the environmental performance of beef jerky and plant-based jerky. The selection was supported by previous studies comparing the environmental performance of beef products with their plant-based alternatives (Heller et al. 2025; Saerens et al. 2021; Saget et al. 2021b; Tang et al. 2024).

2.6 Sensitivity and uncertainty analysis

Sensitivity and uncertainty analyses were conducted to assess the robustness of the results (European Commission 2021). The baseline assessment used the adapted EF 3.1 method. Methodological sensitivity was tested using ReCiPe 2016 (Huijbregts et al. 2017), while scenario sensitivity was assessed by modelling beef jerky produced from beef sourced from France (base model; Beef, round, raw, packed in FR|Chilled|PS| at distribution{FR}U (AGRIBALYSE V3.1 (Koch et al. 2024)) and beef jerky produced from dairy cow meat in Europe (base model; Dairy cow meat, at slaughterhouse {RER} Economic, U (Agri-footprint v6.3 (Durlinger et al. 2014)). An additional sensitivity analysis was conducted to evaluate the influence of database selection on the results.

Uncertainty was evaluated in SimaPro 9.6.0.1 through a Monte Carlo simulation with 1,000 iterations. Because parameter-specific uncertainty ranges were unavailable for the foreground inventories, primary and literature-based inputs including ingredient quantities, extrusion and drying energy requirements, cooling inputs, and other process parameters were treated as fixed values. The analysis therefore propagated the uncertainty distributions embedded in the ecoinvent v3.10 (Wernet et al. 2016), AGRIBALYSE V3.1 (Koch et al. 2024), and Agri-footprint v6.3 (Durlinger et al. 2014) background datasets. Missing data were excluded, while alternative geographical sourcing and database choices were addressed separately through sensitivity analyses. For the comparative Monte Carlo analysis, the difference between the two product systems was calculated for each iteration. The proportion of iterations in which one product showed a lower impact was interpreted as the probability of dominance, and differences were considered robust when this probability was ≥ 95%. Accordingly, no parameter uncertainty was assigned to the modelled foreground processes. This represents a limitation of the uncertainty analysis, as the resulting uncertainty distributions account only for background uncertainty and may therefore underestimate the overall uncertainty of the modelled systems.

3 Results

3.1 Environmental impacts of beef jerky and plant-based jerky

The midpoint results (Table 1) show clear differences in environmental performance between beef jerky and its plant-based analogue. Beef jerky exhibited the highest overall impact, contributing more to 15 out of 16 categories. In key environmental categories, such as climate change, land use, and water use, the impacts of 1 kg of beef jerky were 73.71 kg CO_2 eq, 5593.17 Pt, and 6.53 m^3, respectively. In comparison, the same amount of plant-based jerky contributed 3.65 kg CO_2 eq, −528.09 Pt, and 3.33 m^3 to the respective categories, corresponding to reductions of approximately 95.0% and 49.0% for climate change and water use, respectively. Land use impacts decreased from 5593.17 Pt to −528.09 Pt. Ionising radiation is the only impact category that showed higher impacts than beef jerky. On a mass basis, plant-based jerky showed a 39.7% higher impact in this category. On a protein basis, the relative impact reductions for climate change and water use decreased to 92.9% and 26.8%, whereas land use impacts

Table 1 Results in the midpoint impact categories expressed per functional unit (functional unit: 1 kg of product or 1 kg protein of product, the highest impacts are marked in bold and underlined letters, method - Environmental Footprint (EF) 3.1)

Impact category	Unit	Functional Unit			
		1 kg of product		1 kg protein of product	
		Beef jerky	Plant-based jerky	Beef Jerky	Plant-based jerky
Acidification	mol H+eq	**0.65**	0.03	**1.16**	0.07
Climate change	kg CO_2 eq	**73.71**	3.65	**131.62**	9.36
Ecotoxicity, freshwater	CTUe	**1559.43**	552.80	**2784.69**	1417.43
Particulate matter	disease inc.	$\mathbf{1.06 \cdot 10^{-5}}$	$2.17 \cdot 10^{-7}$	$\mathbf{1.89 \cdot 10^{-5}}$	$5.57 \cdot 10^{-7}$
Eutrophication, marine	kg N eq	**0.58**	0.01	**1.03**	0.04
Eutrophication, freshwater	kg P eq	**0.01**	$1.97 \cdot 10^{-3}$	**0.03**	0.01
Eutrophication, terrestrial	mol N eq	**1.33**	0.09	**2.38**	0.23
Human toxicity, cancer	CTUh	$\mathbf{2.42 \cdot 10^{-8}}$	$8.58 \cdot 10^{-9}$	$\mathbf{4.32 \cdot 10^{-8}}$	$2.20 \cdot 10^{-8}$
Human toxicity, non-cancer	CTUh	$\mathbf{9.47 \cdot 10^{-7}}$	$3.71 \cdot 10^{-7}$	$\mathbf{1.69 \cdot 10^{-6}}$	$9.52 \cdot 10^{-7}$
Ionising radiation	kBq U-235 eq	1.26	**1.76**	2.25	**4.52**
Land use	Pt	**5593.17**	−528.09	**9987.80**	−1354.07
Ozone depletion	kg CFC11 eq	$\mathbf{1.95 \cdot 10^{-6}}$	$2.21 \cdot 10^{-7}$	$\mathbf{3.48 \cdot 10^{-6}}$	$5.66 \cdot 10^{-7}$
Photochemical ozone formation	kg NMVOC eq	**0.15**	0.01	**0.27**	0.03
Resource use, fossils	MJ	**212.11**	81.39	**378.77**	208.68
Resource use, minerals and metals	kg Sb eq	$\mathbf{1.14 \cdot 10^{-4}}$	$2.28 \cdot 10^{-5}$	$\mathbf{2.04 \cdot 10^{-4}}$	$5.85 \cdot 10^{-5}$
Water use	m^3 depriv.	**6.53**	3.33	**11.66**	8.53

further decreased for plant-based jerky from −528.09 Pt to −1354.07 Pt, widening the gap between both products.

The single score (Fig. S 1, SM 1) presents the aggregated environmental impact of the products, facilitating comparison and identifying the most relevant impact categories. Although weighting approaches are useful for communicating impact results, they must be evaluated critically, as they primarily reflect political, cultural, or personal preferences rather than scientific principles (Sala et al. 2018). On a product basis, beef jerky reached 7.66 mPt, whereas plant-based jerky accounted for 0.63 mPt (8.2% of the meat-based impact). On a protein basis, plant-based jerky scored 1.62 mPt, corresponding to 11.8% of beef jerky's total score (13.68 mPt), Therefore the overall environmental impact of plant-based jerky was lower across both functional units. For beef jerky, six impact categories contributed 80% to the total impact: climate change (26.8%), particulate matter (20.8%), marine eutrophication (11.4%), acidification (9.4%), land use (7.1%), and freshwater ecotoxicity (6.9%). For plant-based jerky, the main contributors were freshwater ecotoxicity (29.7%), fossil resource use (16.5%), climate change (16.1%), human toxicity (non-cancer) (8.4%), freshwater eutrophication (5.4%), and particulate matter (5.2%).

3.2 Relative contributions of unit processes

The raw material production stage of beef jerky showed the highest impacts across all impact categories among unit processes (Fig. 3). In addition, the drying and packing stages contributed notably to ionising radiation and fossil resource use. Drying accounted for 23.5% and 5.1% of these impacts, respectively, while packaging accounted for 5.1% and 10.0%, respectively. Slicing, marinating, and cutting showed minor significance. Accordingly, ingredient production constituted the main hotspot in the beef jerky production system.

The relative contributions to the impact categories of plant-based jerky (Fig. 4) were more evenly distributed across production stages compared to beef jerky. As with beef jerky, raw material production remained the dominant processing stage, contributing more than 40% to most impact categories, except land use. Besides raw material production, the cooling and packaging stages were also identified as hotspots in key impact categories, including climate change and water use. For ionising radiation, raw material production was the dominant contributor, at 66.39%. This was followed by cooling (16.74%), extrusion (6.35%), drying (6.35%), and packing (3.64%).

3.3 Relative contributions of raw materials

Raw material acquisition accounted for a significant proportion of the overall environmental impact of both products. Therefore, the cumulative contributions of raw materials within each ingredient category were analysed in greater detail (Figs. 5 and 6).

Beef meat was identified as the dominant contributor to the environmental impacts of the beef jerky formulation, accounting for more than 88% of all impact categories (Fig. 5). In the categories ionising radiation, water use, and resource use (fossils), contributions from spices and additives corresponded to 10.3%, 7.7%, and 4.1%. Other

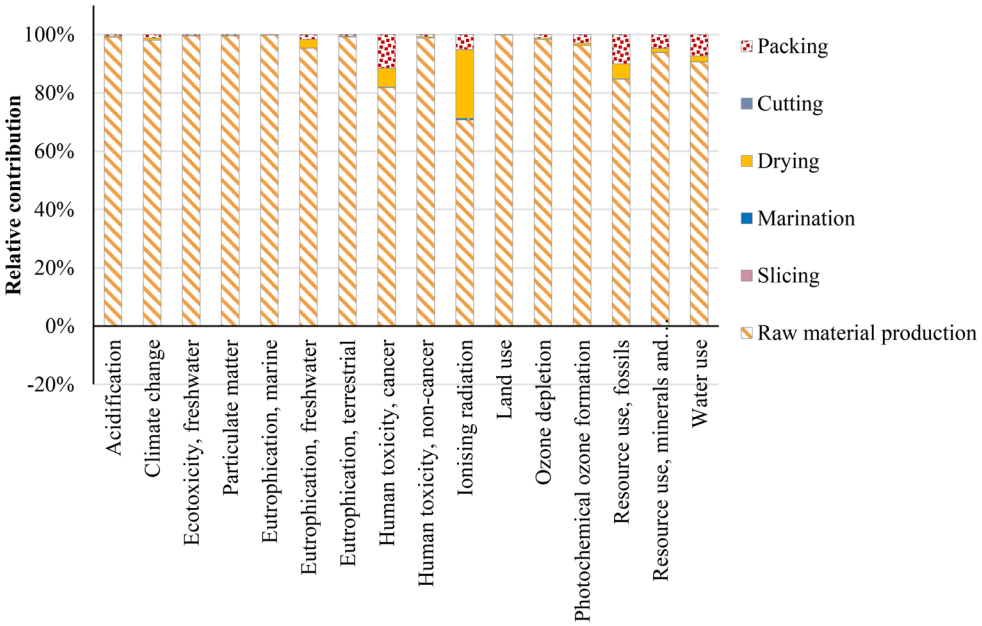


Fig. 3 Relative environmental impact (%) of beef jerky production stages (functional unit- 1 kg product, method- Environmental Footprint (EF) 3.1)

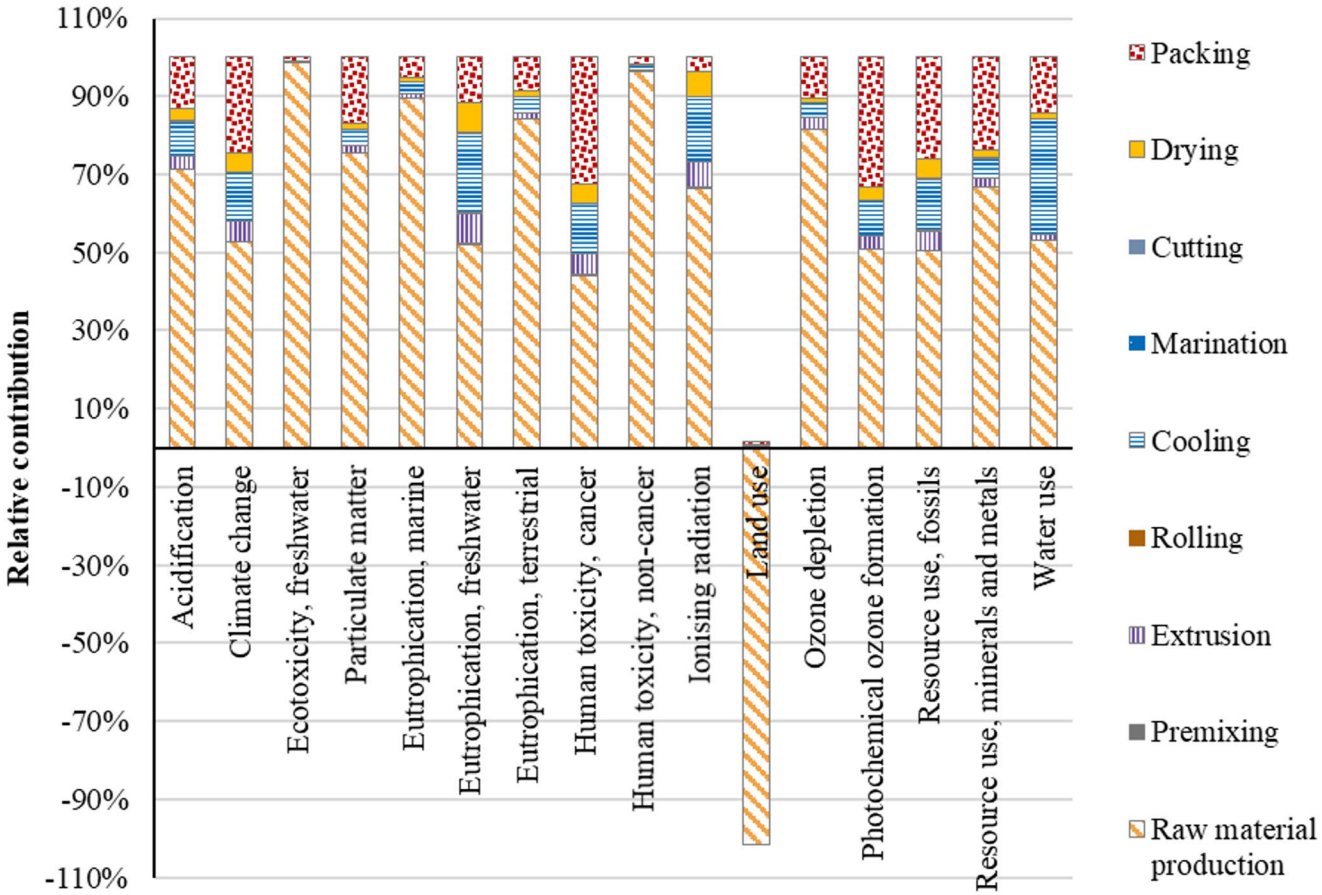


Fig. 4 Relative environmental impact (%) of plant-based jerky production (functional unit − 1 kg, method - Environmental Footprint (EF) 3.1)

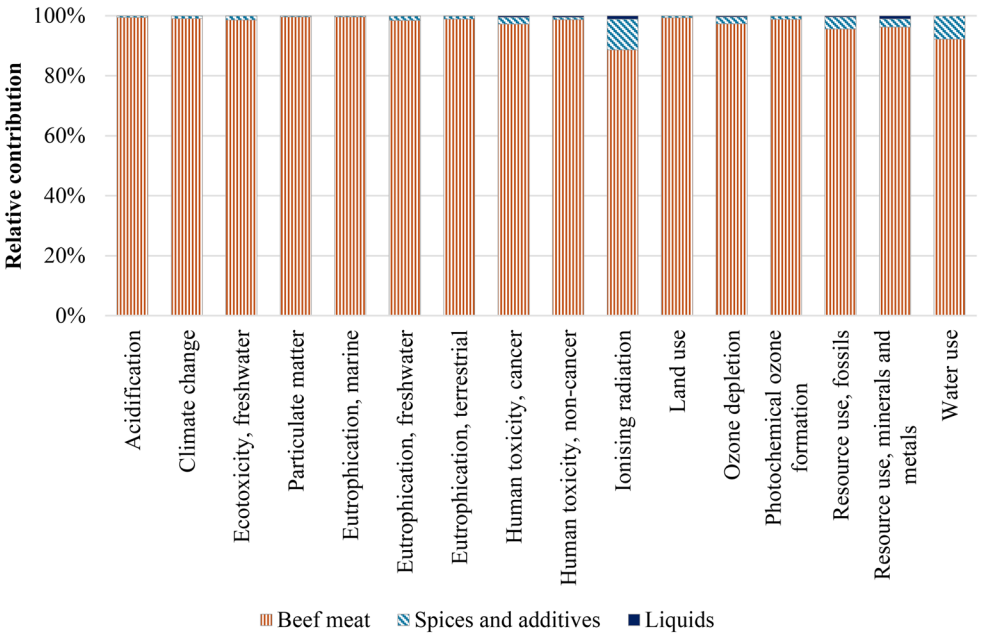


Fig. 5 Relative contributions (%) to impact categories of beef jerky components (method - Environmental Footprint (EF) 3.1)

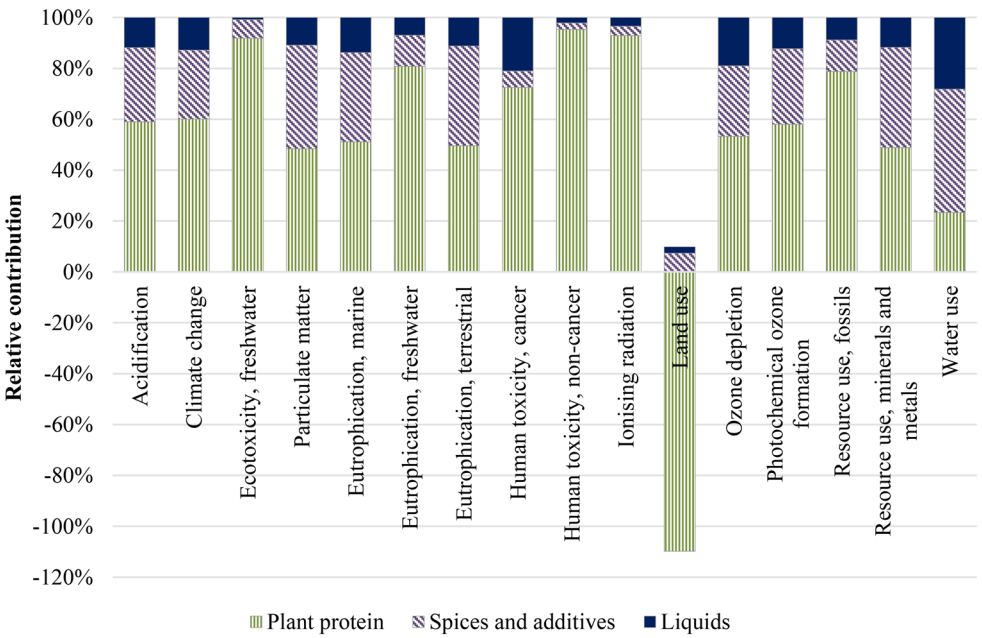


Fig. 6 Relative contributions (%) to impact categories of plant-based jerky components (method - Environmental Footprint (EF) 3.1)

impacts from spices and liquids were only of minor significance for the environmental performance.

For plant-based jerky, the combined contribution of plant proteins exceeded 50% in nearly all impact categories (Fig. 6). Exceptions were observed for land use due to negative land use impacts, terrestrial eutrophication (49.6%), mineral and metal resource use (48.9%), particulate matter (48.4%), and water use (23.4%). The water use category was mainly dominated by spices and liquids, with a share of 48.5% and 28.1%, respectively, of its total impact.

3.4 Environmental impacts of protein sources

As shown in Fig. 7, beef had the highest single-score impact among the assessed protein sources (3.31 mPt), with climate change contributing the largest share (26.8%). Among the plant proteins, pea protein isolate showed the highest score (1.29 mPt), mainly driven by freshwater ecotoxicity (63.5%), followed by sunflower seed protein concentrate (0.54 mPt), for which fossil resource use was the dominant category (32.2%). Pumpkin seed flour had a substantially lower score (0.15 mPt), while faba bean protein concentrate showed a negative score (−0.26 mPt).

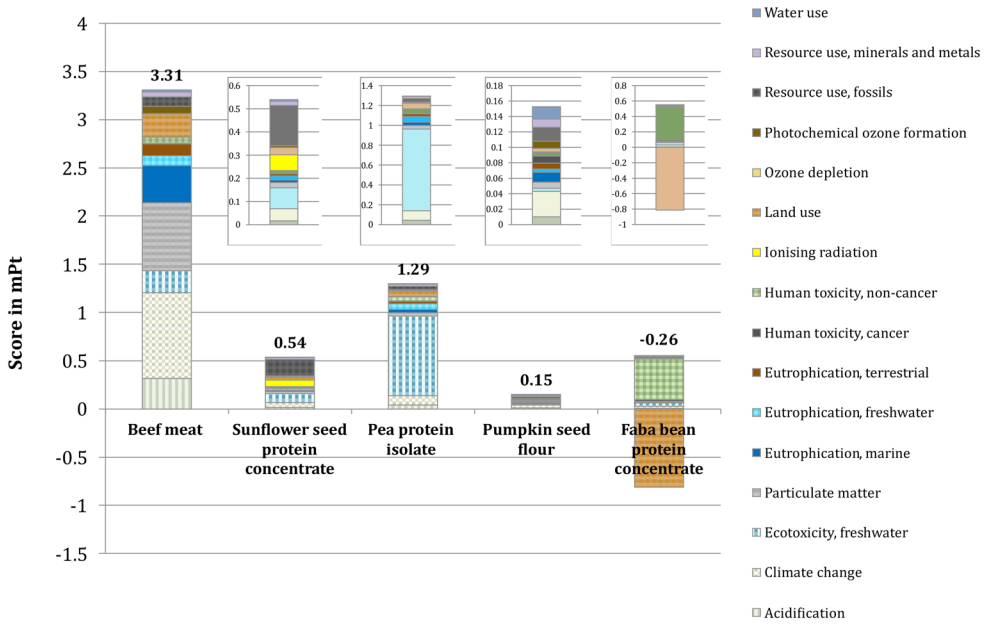


Fig. 7 Single score (integrated environmental impact) comparison of the different protein sources, including beef meat, sunflower seed protein concentrate, pea protein isolate, pumpkin seed meal, faba bean protein concentrate (functional unit−1 kg of product, method - Environmental Footprint (EF) 3.1, unit - millipoint (mPt))

3.5 Sensitivity analysis

To assess the robustness of the results to the choice of method, the environmental impacts of beef jerky and plant-based jerky were recalculated using the ReCiPe 2016 method as an alternative to EF 3.1. Across the ReCiPe 2016 results (Tab. S 2, SM 1), beef jerky showed higher impacts in 17 out of 18 impact categories, except for ionising radiation. However, some category-specific results differed between methods. For instance, EF 3.1 produced negative values for land use in the plant-based jerky system, whereas ReCiPe 2016 reported positive impacts.

In addition to selecting an alternative method, the robustness of the results was evaluated by assessing the influence of beef origin and cattle farming system. The scenario sensitivity analysis (Tab S 3, SM 1) demonstrated that changing the beef origin altered the magnitude of several impacts but did not change the overall comparison between products. French beef jerky had lower impacts than Irish beef jerky in 11 of 16 categories, including climate change, freshwater ecotoxicity, particulate matter, marine eutrophication, land use, and fossil resource use. However, higher impacts were observed for acidification, terrestrial eutrophication, cancer-related human toxicity, ionising radiation, and water use. A substantial improvement was observed when Irish beef was replaced with dairy cow meat {RER} (Agri-footprint v6.3 (Durlinger et al. 2014), showing lower impacts in each category. In the key impact categories climate change, land use, and water use, impacts were reduced by 69.5%, 72.9%, and 3.8%, respectively. However, plant-based jerky remained the lowest-impact option in nearly all categories, confirming the robustness of the main conclusions.

Given the substantial contribution of sunflower protein concentrate to ionising radiation, an additional sensitivity analysis was conducted by remodelling sunflower protein concentrate using an adapted soy protein concentrate inventory from Agri-footprint v6.3 (Durlinger et al. 2014), instead of the AGRIBALYSE v3.1 dataset (Koch et al. 2024) used in the baseline model. The two datasets reflect different protein extraction processes and, therefore, differ in energy requirements and other inventory inputs. The resulting impact assessment is presented in Tab S 4, SM 1. Although this change did not alter the overall comparison between beef and plant-based jerky, the adapted model showed notable improvements in some impact categories, particularly ionising radiation and resource use (fossils). Compared with the base model of plant-based jerky, impacts in these categories were 56.8% and 26.0% lower.

3.6 Uncertainty analysis

The uncertainty analysis (Tab. S 5, SM1) showed that most impact categories exhibited moderate uncertainty, with coefficients of variation (CV) below the 20% threshold. Compared with beef jerky, plant-based jerky showed greater variability in impact results. Higher CV values were observed for the impact categories water use, ionising radiation, and toxic-related categories for both products (Tab. S 5, SM1).

To confirm the overall result, the certainty level of difference between the two jerky products was evaluated in each impact category (Fig. S 2, SM1). In 13 of 16 midpoint impact categories, the difference between the two products was statistically significant, exceeding the 95% confidence threshold. In the categories of human toxicity (cancer),

human-toxicity (non-cancer), and water use the confidence threshold could not be reached. The level of certainty that the production of plant-based jerky has less impact on those categories than beef jerky accounted for 91.9%, 68.8%, and 64.4%, respectively.

4 Discussion

4.1 General discussion

The comparison of the overall environmental performance of beef jerky and plant-based jerky showed that beef jerky had the highest overall impact, with higher contributions in 15 of 16 categories. This result is consistent with existing comparisons of meat and plant-based products (Heller et al. 2025; Saerens et al. 2021; Saget et al. 2021b; Tang et al. 2024). However, the climate change impact of 1 kg of beef jerky (73.71 kg CO_2eq) appeared to be higher compared to other beef-based food products investigated in previous studies (Saerens et al. 2021; Saget et al. 2021a, b). For instance, the climate change impact of beef meatballs ranged from 33 to 46 kg CO_2eq without considering the carbon opportunity cost (COC) factor (Saget et al. 2021a). This difference can largely be explained by differences in product composition and processing. Meatballs are high-moisture products, whereas beef jerky is dried and therefore requires more raw meat per kilogram of final product. In the present study, 2.25 kg of fresh beef is required to produce 1 kg of beef jerky, indicating a minimum 2.25-fold increase in environmental impacts compared with raw beef and related fresh products. According to Saget et al. (2021a), only 0.70 kg of raw beef is required to produce 1 kg of meatball. This comparison evidence that the result of the present study is comparable to other beef meat and related product environmental assessments. Furthermore, the present study has identified beef meat production as the main environmental hotspot within the beef jerky production system (Fig. 5). Similar trends have been reported by Saget et al. (2021a), with beef production contributing at least 76% to key impact categories of beef meatballs. Considering that climate change was identified as the dominant impact category of beef meat production according to the single score assessment (26.8%) (Fig. 7), the high quantity of beef required for dried jerky was the main factor driving its climate change impact.

Plant-based jerky performed better than beef jerky in nearly all impact categories. Substantial lower impacts were achieved in key categories, such as climate change, land use, and water use, showing reductions of 95% for climate change and of 49% for water use. Land use impacts decreased from 5593.17 Pt to −528.09 Pt. While meat alternatives commonly reduce land use (Heller et al. 2025; Saerens et al. 2021; Tang et al. 2024), negative values are rarely reported. The negative land-use contribution was attributed to the faba bean protein concentrate dataset. An elementary-flow contribution analysis showed that it was associated primarily with the flows "Transformation, from unspecified" and "Transformation, to annual crop, irrigated." The inventory contains positive quantities of land occupation and transformation flows; therefore, the negative result does not originate from a negative inventory input or an avoided-product credit. In EF 3.1, land-use impacts are characterized using a soil-quality index based on the LANCA framework (Andreasi Bassi et al. 2023; Fazio et al. 2018). The method assigns characterization factors to land occupation and transformation flows according to their effects on soil functions, including biotic production, erosion resistance, groundwater regeneration, and mechanical and physicochemical filtration. The net negative value therefore results from the combination of the land-transformation flows contained in the selected background dataset and their corresponding EF 3.1 characterization factors. It is not an explicit credit for biological nitrogen fixation, reduced fertilizer use, or improved soil fertility, as these crop-specific agronomic effects are not directly represented by the EF 3.1 land-use characterization model. Consequently, the negative value should be interpreted as a dataset- and characterization-model-dependent result rather than as evidence of an absolute environmental benefit from faba bean cultivation.

Ionising radiation was the only impact category in which plant-based jerky performed worse than beef jerky, with a 39.7% higher impact on a mass basis. Raw material production was the main contributor, accounting for 66.4% of the total impact. Among the raw materials, sunflower protein concentrate contributed most, accounting for 89.3% of the raw material-related impact (Fig. S 3, SM1). A detailed contribution analysis showed that 98.8% of the ionising radiation impact of sunflower protein concentrate originated from electricity consumption during protein extraction (Fig. S 4, SM1). According to the inventory data (SM2), sunflower protein concentrate production required more electricity than other plant protein processes because it used wet extraction. This is consistent with previous findings that plant protein production can require substantial energy and water inputs (Detzel et al. 2021). Furthermore, sunflower protein concentrate and pea protein isolate represented the largest share of the plant-based jerky formulation. Heat and electricity consumptions are closely linked to impacts in categories such as fossil resource use and ionising radiation (Bonales et al. 2024), which explains the relatively high impact of sunflower protein concentrate. Beyond raw material production, ionising radiation impacts were associated with cooling (16.74%), extrusion (6.35%), drying

(6.35%), and packaging (3.64%). These stages all depend on electricity, explaining their contribution to this energy-related impact category. The relatively high cooling impact was mainly linked to using liquid nitrogen as a cryogenic fluid. Its production involves energy-intensive air compression and fractional distillation, along together with transport and transfer losses (Cytiva 2020). These results indicate that improved cooling efficiency and electricity management could further reduce the environmental impacts of plant-based jerky.

Moreover, water-use reductions were smaller than those reported in previous studies for meat analogues, which often range from 92% to 97% (Bonales et al. 2024; Heller et al. 2025). However, the meat analogues examined so far often relied on simple recipes consisting of one or two plant protein sources (Bonales et al. 2024; Heller et al. 2025). In contrast, plant-based jerky formulation appears more complex, containing 16 different ingredients (Tab. S 1, SM1). As with beef jerky, raw material production had the largest impact on the environmental performance of plant-based jerky, with plant proteins exceeding 50% in nearly all impact categories. However, spices and additives were most impactful on water use (48.5%), even though they represented a minor part of the formulation (Tab. S 1, SM1). Likewise, Bonales et al. (2024) reported that complementary ingredients, including oils, starches, and spices, can significantly influence certain environmental impacts of plant-based product formulations and should be carefully evaluated during product development. In the present study, spices' high contribution to water use was linked to their cultivation. Additionally, tomato paste production contributed notably to water use. In the tomato paste model, tomatoes were assumed to be cultivated in greenhouses, which require substantial water inputs for irrigation and for producing plastic tunnel materials.

The functional unit did not affect the overall result. However, on a protein basis, the relative impact reductions for plant-based jerky in climate change and water use accounted for only 92.9% and 27.0%, whereas the gap in land-use impacts increased considerably. These differences arise from the lower protein content of the plant-based analogue. 100 g of beef jerky provides 56 g of protein, compared with 39 g in the plant-based alternative. To achieve an equivalent protein intake, consumers must eat 44 g more plant-based jerky, resulting in higher impacts.

4.2 Sensitivity analysis

The choice of LCIA method can influence LCA results and their interpretation because characterization models, factors, units, and impact pathways differ across methods (Owsianiak et al. 2014). In the present study, plant-based jerky remained environmentally preferable under both EF 3.1 and ReCiPe 2016, indicating that the main conclusion was robust to methodological choice. Nevertheless, category-specific differences were observed. For example, EF 3.1 produced a negative land-use result for plant-based jerky, whereas ReCiPe 2016 reported a positive value, reflecting differences in land-use modelling and characterization rather than a change in the underlying inventory.

The additional scenario analyses showed that geographical sourcing and database selection affected the magnitude of several impacts. Replacing Irish beef with French beef changed results across multiple categories. While most impact categories improved, 5 of 16 categories showed higher impacts than Irish beef. This result aligns with findings by Kokemohr et al. (2022), who reported variations in the environmental performance among cattle production systems of different origins. In addition, the scenario analysis showed that replacing Irish beef with dairy cow meat reduced environmental impacts across all impact categories. This difference largely stems from the allocation procedures used in the underlying datasets. In the dairy production model, approximately 92% of the environmental burden is allocated to milk, while only about 6% is assigned to the dairy cow and around 2% to the calf. Consequently, when the dairy cow dataset is used as input to the dairy cow meat production model, only the relatively small share initially allocated to the cow carries forward. At the subsequent meat-processing stage, approximately 95% of this share is allocated to dairy cow meat, with the remainder assigned to other co-products. In contrast, the beef cattle production model allocates the entire environmental burden to the beef animal, of which approximately 97% is subsequently attributed to beef meat. Therefore, the markedly lower impacts of dairy cow meat primarily reflect the allocation structure applied within the dairy system. This finding is consistent with Cederberg and Stadig (2003), who demonstrated that allocation choices can strongly influence the environmental impacts attributed to dairy and meat products. Nevertheless, plant-based jerky remained the option with the lowest overall environmental impacts. Finally, remodelling of sunflower protein concentrate, using an inventory adapted from Agri-footprint v6.3 (Durlinger et al. 2014), reduced impacts particularly for ionising radiation and resource use. However, none of the three sensitivity assessments altered the overall ranking of the products. Together, these findings demonstrate that, although individual impact results remain sensitive to methodological and inventory choices, the principal conclusion of the study is robust.

4.3 Uncertainty analysis

The comparative uncertainty analysis showed that the difference between the two products exceeded the 95% confidence threshold in most impact categories, supporting the conclusion that plant-based jerky generally had lower environmental impacts than beef jerky, except for ionising radiation. The 95% threshold was not reached for human toxicity (cancer), human toxicity (non-cancer), and water use, indicating that the differences between products were less robust in these categories. These results should, however, be interpreted in relation to the scope of the Monte Carlo analysis, which primarily propagated uncertainty distributions embedded in the selected background databases, while treating foreground inventory values and literature-based assumptions as fixed inputs. The comparatively greater uncertainty observed for plant-based jerky may therefore reflect its reliance on several plant protein ingredients and a larger number of upstream processing datasets rather than inherently greater uncertainty in the product itself. In addition, water use, ionising radiation, and toxicity-related categories are known to exhibit comparatively high variability because of limitations in their characterization models within the EF framework (European Commission 2021). Toxicity-related indicators were considered insufficiently robust for external communication during the EF pilot phase (Sala et al. 2018). Although updated characterization factors have addressed some limitations (Sala et al. 2018), substantial uncertainty remains. Accordingly, conclusions for these categories should be interpreted cautiously, whereas the overall comparative ranking of the products was robust across most impact categories.

4.4 Limitations and future research directions

Several methodological limitations should be considered when interpreting the findings of this study. Due to the lack of primary data for beef jerky production, the model relied mainly on secondary literature, while the plant-based jerky model was based mostly on data from a single manufacturer. Therefore, the results may not be representative of all products available on the market. In addition, both systems required assumptions where primary data were unavailable, particularly for drying energy demand, which was estimated using simplified calculations and did not fully account for changes in product composition during moisture removal. The absence of comparable LCA studies on jerky products also limited direct benchmarking of the results.

A further source of uncertainty arises from combining different LCI databases. Although dataset selection prioritised technological, geographical, and product representativeness, differences remain among ecoinvent, AGRIBALYSE, and Agri-footprint in activity data, agricultural emission modelling, allocation procedures, land-use modelling, and the treatment of infrastructure and capital goods (Corrado et al. 2018). Agri-footprint v6.3 (Durlinger et al. 2014) partly addresses the infrastructure-related limitations identified in earlier versions of Agri-footprint; however, it still does not consistently represent capital goods across all processes. Available evidence indicates that infrastructure and capital goods often contribute relatively little to climate-change impacts and other operationally dominated categories, but may be more relevant for resource-use and toxicity-related indicators (Mahlan et al. 2025). Consequently, differences in database structure are unlikely to dominate all impact categories but remain an additional source of methodological uncertainty.

Future studies would benefit from incorporating primary production data from multiple manufacturers and more detailed measurements of processing requirements to improve model accuracy and representativeness. In addition, comparisons based on protein supply would benefit from greater consideration of protein quality and human protein utilization. Animal and plant-based proteins differ in digestibility, bioavailability, and essential amino acid composition, with plant proteins generally showing lower digestibility and limitations in some essential amino acids (Berrazaga et al. 2019; Iqbal 2022). Incorporating these nutritional differences into future LCA comparisons could provide a more comprehensive assessment of the environmental performance of animal- and plant-based protein products.

5 Conclusions

The present study compared the environmental performance of beef jerky and a plant-based alternative using life cycle assessment. Across both functional units (1 kg of product and 1 kg of protein), the plant-based jerky showed lower impacts in 15 out of 16 assessed impact categories, while beef jerky performed better only in the category ionising radiation. On a product-mass basis, the plant-based alternative showed substantially lower impacts for climate change, land use, and water use, including reductions of approximately 95% and 49% for climate change and water use, respectively. The land-use impact result for the plant-based product was negative. This negative value was rather attributed to the selected characterization method and data set for faba bean protein concentrate than to an unambiguous absolute environmental benefit.

Protein fractionation processes and the inclusion of complementary ingredients, such as spices, strongly influenced the environmental performance of plant-based jerky,

increasing impacts in some categories. These findings highlight the importance of considering both processing intensity and formulation choices when developing plant-based products. As most existing LCA studies focus on meat alternatives based on a single plant protein source, future research would benefit from evaluating the combined use of different plant proteins that may improve amino acid balance and nutritional quality while also assessing their environmental implications. Further integration of nutritional aspects, including protein digestibility, bioavailability, and human protein utilization, could provide a more comprehensive basis for comparing plant- and animal-based foods.

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Data availability All data collected and used for this study are available in the supplementary materials.

Declarations

Generative AI and AI-assisted technologies in the writing process During the preparation of this work the author(s) used ChatGPT in order to improve the language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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